

Video Slides W3: Demand, Supply and Competitive Markets

ECO 101: Principles of Microeconomics

Prof. Freitas, University of Toronto

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Perfect Competition

Topic: Demand, Supply and Competitive Markets

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Video Learning Objectives

- Describe the assumptions behind perfectly competitive markets.
- Clarify some topic jargon.
 - Quantity demanded vs. demand.
 - *Ceteris paribus* i.e. all else held equal.
 - Equations vs. discrete values.

Market

- Market: space where buyers and sellers can trade.
- What are market rules?

Assumptions: Perfect Competition

- Buyers and Sellers are “small” relative to the market, i.e. price takers.
- Consumers treat all goods as identical, no transactions costs.
- Full information.
- Free entry and exit.
- No externalities.

Perfectly Competitive Market

- Useful for economic insights, channels and as a benchmark.

Graphing Conventions

- Always label your axes.



Jargon

- Useful as a way to communicate information efficiently.
- We expect you to explain the economic intuition behind your analysis and not just use jargon.

Quantity demanded vs. Demand

- When the **price of a good changes**, all else held fixed, the associated change in the number of units bought is the change in **quantity demanded**.
- Later: movement **along** the curve.

Quantity demanded vs. Demand

- If the quantity people buy at the **same price** changes, we have a **change in demand**
- Later: movement **of** the curve.

Ceteris Paribus

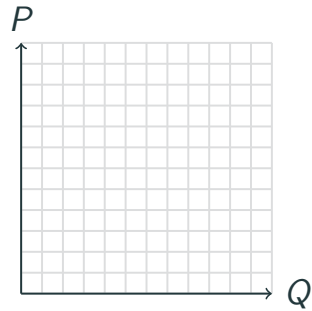
- “All else held fixed”
- Assuming all factors except for one stay the same allows us to focus on the impact of the chosen factor via the model.

Discrete vs. Continuous Data

- Data can be either discrete or continuous.
- Use the information you are given and do not convert it into your most preferred format.

Discrete Data

P	\$10	\$7	\$6	\$3	\$1
Q^D	2	4	6	8	10



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Continuous Data

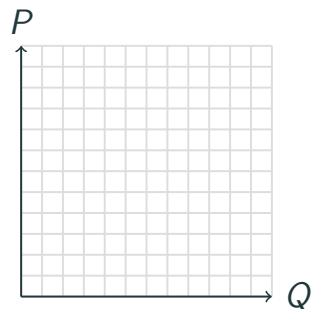
- Data approximated by an equation e.g.
 $Q^D(P) = 10 - \frac{P}{2}$
- Can be re-written as $P(Q^D) =$

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Continuous Data

$$Q^D(P) = 10 - \frac{P}{2}$$



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Summary

- Perfectly Competitive Market assumptions are unrealistic, but they are useful to start with.
- Focus for now: buyers and seller are price-takers for identical goods, no transactions costs.
- Understand ECO101 jargon, it is particularly useful for multiple choice questions, but keep focus always on economic intuition.

Individual Demand

Topic: Demand, Supply and Competitive Markets

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Video Learning Objectives

- Explain how to determine a relationship between quantity demanded and price for an individual, represent and interpret it in equation form and on a graph.
- Explain how a demand curve is related to Marginal Willingness to Pay.

Perfectly Competitive Market

- Market is made up of individuals so to get to market demand we start with the individual demand.
- Buyers (and Sellers) are “small” relative to the market.
 - Each buyer can buy as many units as they want without affecting market price.
 - Buyers react to prices.

Recall

MB: marginal benefit, MC: marginal Cost

- Buy extra unit if $MB \geq MC$.
- $MB =$ Marginal Willingness to Pay
- $MC =$ Explicit cost i.e. market price

Quantity demanded at a given price?

MB: Marginal Benefit, MC: Marginal Cost, MWTP: Marginal Willingness to Pay.

- How many units to buy at a given price?
- Use insight: buy unit the $MB(i.e. MWTP) = MC$ for the last unit.
 - Implicit assumptions?

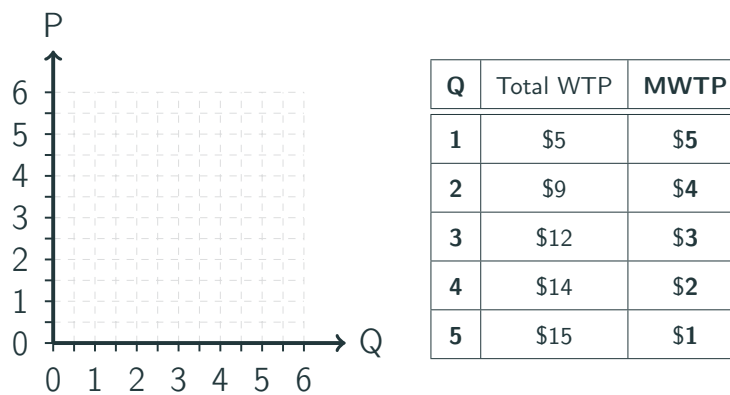
Graphing Demand

Thought experiment:

- Pick a price.
- Find how many units the consumer is willing to buy at that price i.e. how many units pass the $MWTP \geq P$ bar?
- Change price, all else held fixed.
- Repeat until you have your demand curve.

Graphing Demand

MWTP = Marginal Willingness To Pay



Law of Demand

- As the price of a good decreases, all else held fixed, consumers want to buy a larger quantity of the good.
- Why?
 - Buy additional unit if $MB \geq MC$ (=price).
 - So as price decreases...

Summary

- An individual's demand curve plots quantity demanded for any price.
- It represents the consumer's Marginal Willingness to Pay for every unit.
- Remember as we move along the curve we vary price, all else held fixed

Changes in Demand

Topic: Demand, Supply and Competitive Markets

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Video Learning Objectives

- Differentiate between movements of vs. along the curve
- Identify factors that cause a shift of the demand curve and the direction of change.
- Classify goods into substitutes, complements, inferior and normal goods.

Demand

- Decisions based on MWTP vs. Price
- Recall from earlier module MWTP vs. Price is just a reworked version of Benefits vs. Opp. Cost.
- A demand curve changes just price, all else held fixed.

“All else held fixed”

What non-price related things affect how much a person wants to buy at a given price?

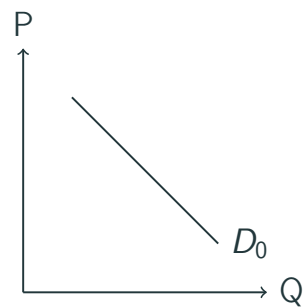
- Tastes and preferences
- Prices of related goods.
- Income.
- Expected prices in future.
- (Market) No. of consumers in the market

Changes in the prices of related goods

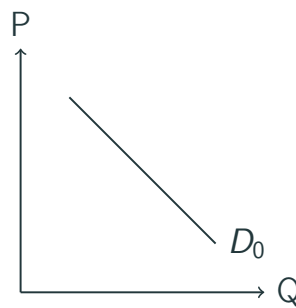
- Substitutes: Good B can be substituted for good A.
- Complements: We get more value from good C if good A is consumed as well.

Changes in the prices of related goods

Substitutes $P_{\text{related}} \uparrow$

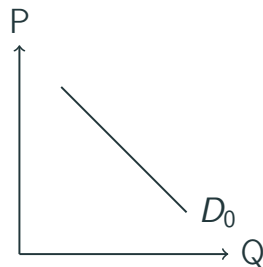


Complements $P_{\text{related}} \uparrow$



Expected Future prices

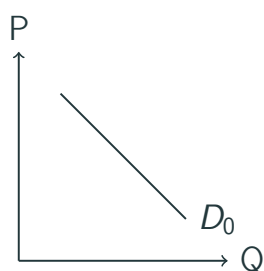
If consumers are able to **substitute** buying the good today for buying the good tomorrow, then a change in future prices of the good...



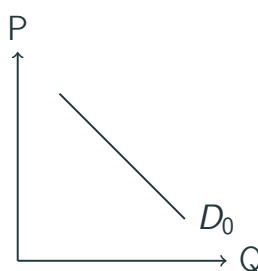
Changes in income (or wealth)

- Normal: As income $\uparrow$, demand for good $\uparrow$.
- Inferior: As income $\uparrow$, demand for good $\downarrow$.

Normal



Inferior



Summary

- When factors in the “all else constant” part have changed, we have a change in demand.
- Always relate the change in decisions to changes in the MWTP to understand the economics
- How demand shifts has useful information
 - normal vs. inferior, substitutes vs. complements.

Individual Supply

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Video Learning Objectives

- Explain how to determine a relationship between quantity supplied and price for a firm, represent and interpret it in equation form and on a graph.
- Explain how a supply curve is related to Marginal cost.

Seller are Price-takers

Buyers and Sellers are “small” relative to the market

- Each seller can sell as many units as they want without affecting market price.
- Sellers **react** to prices, they do not set prices.

Recall

MB: marginal benefit, MC: marginal Cost

- Sell extra unit if $MB \geq MC$.
- $MB =$ market price.
- $MC =$ (opp.) cost of producing an extra unit.

Quantity Supplied

MB: marginal benefit, MC: marginal Cost.

- For a firm, how many units to sell at a given price?
- Use insight: buy unit the $MB(i : e. \text{ price}) = MC$ for the last unit.

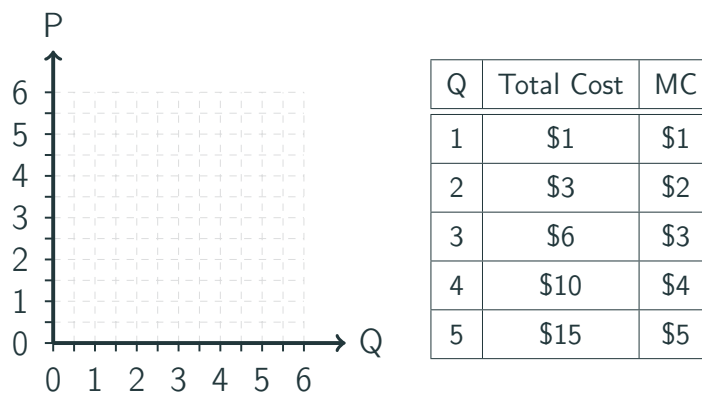
Graphing Supply

Thought experiment:

- Pick a price.
- Find how many units the producer is willing to supply at that price i.e. how many units pass the $P \geq MC$ bar?
- Change price, all else held fixed.
- Repeat until you have your supply curve.

Graphing Supply

MC: Marginal Cost, TC: Total cost



Graphing the Supply Curve

- A firm's supply curve plots quantity supplied for any price.
- It represents the firm's Marginal Costs.

So far in the thought experiment, varied price, all else held fixed

Shape of the Supply Curve

- As the price of a good increases, all else held fixed, does the firm want to sell a larger quantity of the good?
- Why?
 - Sell additional unit if $MB (=price) \geq MC$.
 - So as price increases...

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Summary

- An individual firm's supply curve plots quantity supplied for any price.
- It represents the firm's Marginal Cost curve.
- Remember as we move along the curve we vary price, all else held fixed and firms react to prices, they do not set them.

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Changes in Supply

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Video Learning Objectives

- Movements of vs. along the curve
- Identify factors that cause a shift of the supply curve and the direction of change.

Supply

- Decisions based on Price vs. MC
- A supply curve changes just price, all else held fixed.

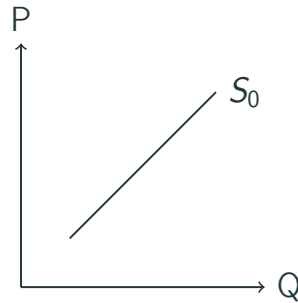
“All else held fixed”

What non-price related things affect how much a person wants to sell at a given price?

- Price of related goods (substitutes in production, complements in production).
- Technology.
- Price of inputs
- Expectations of future costs/prices.
- (Market) Number of sellers

Changes in Technology

- Technology improvement allows more output to be produced using the same inputs



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Changes in the prices of related goods

Production Substitutes:

- B production $\uparrow \rightarrow$ resources for A $\downarrow$.

Production Complements:

- B production $\uparrow \rightarrow$ A production $\uparrow$.

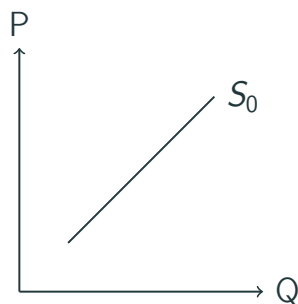
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Changes in the prices of related goods

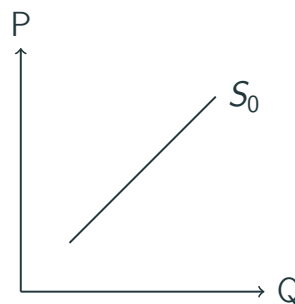
Production Substitutes

$P_{\text{related}} \uparrow$



Production Complements

$P_{\text{related}} \uparrow$



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Summary

- For a given quantity, the point on a supply curve corresponding to that quantity is the Marginal Cost for the last unit.
- For a movement of the curve, factors in the “all else constant” part have changed.
 - Use the direction to classify complements or substitutes in production.

Market Equilibrium

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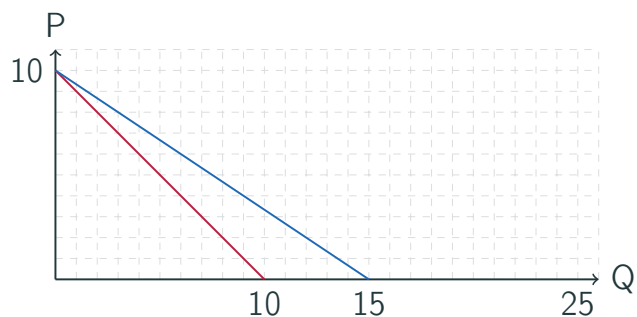
- Summing up individual demand/supply to get market demand/supply
- Calculate and graphically identify market clearing price and equilibrium quantity.
- Explain why equilibrium is reached in a market economy and the role price plays in this process.
- Given change, identify impact on market or given observation, identify what could have caused it.

Individual → Market Demand

- Idea: for a given price, sum up individual demands.
- Math: $\sum_{i=1}^n Q_i(p)$.
- Graphically: Would need to HORIZONTALLY sum up individual demands.

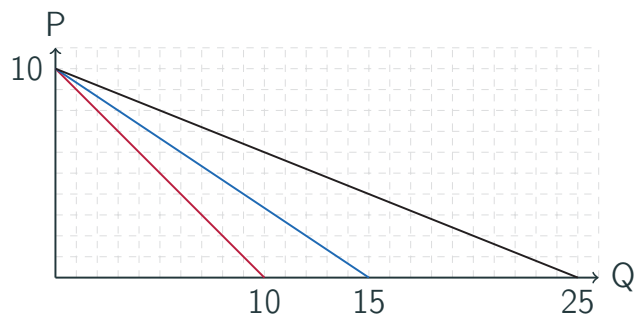
Individual → Market Demand

$$Q_A = 10 - P \text{ and } Q_B = 15 - \frac{3}{2}P$$



Individual → Market Demand

$$Q_A = 10 - P \text{ and } Q_B = 15 - \frac{3}{2}P$$

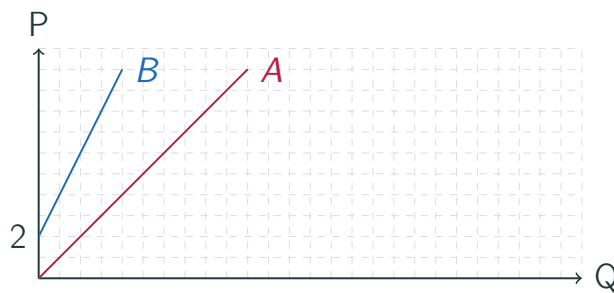


Firm → Market Supply

- Idea: for a given price, sum up firm supply.
- Math: $\sum_{i=1}^n Q_i(p)$.
- Graphically: Would need to HORIZONTALLY sum up firm supplies.

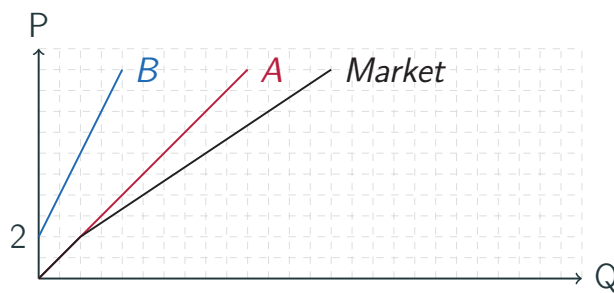
Firm → Market Supply

$$P = Q_A \text{ and } P = 2 + 2Q_B$$



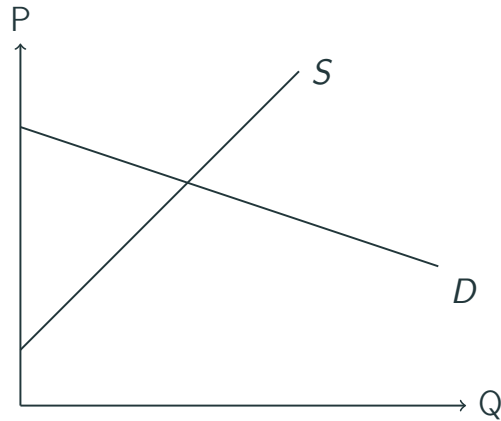
Firm → Market Supply

$$P = Q_A \text{ and } P = 2 + 2Q_B$$



Market Equilibrium

Example: $Q^D = 30 - 3P$ and $Q^S = P - 2$



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Market Equilibrium

- Model with our assumptions predicts that individuals responding to prices will lead us to a stable price and quantity bought/sold.
- In equilibrium quantity demanded $Q^D = Q^S$, quantity sold.
 - graph: intersection of demand and supply
 - math: solve for P^* where $Q^D(P^*) = Q^S(P^*)$

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Shifting Curves

What happens to the market when things change?

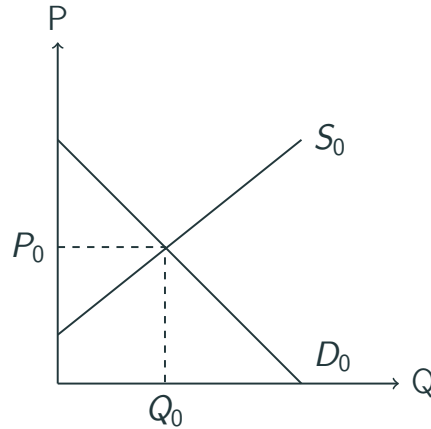
- Identify which curves change and how.
- Check how the change impacts market price and quantity sold.

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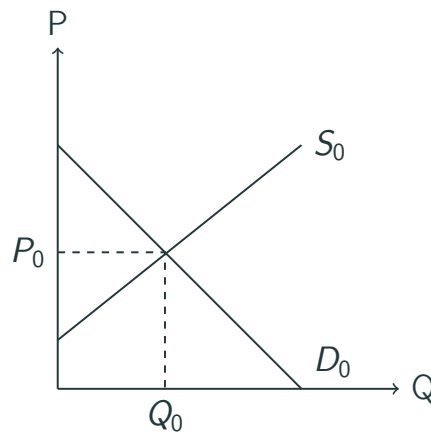
Shifting Supply Out

Example: Technology improvement reduces cost of production



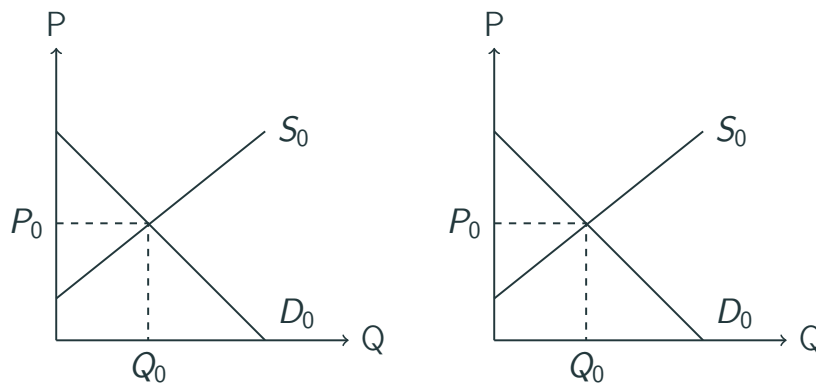
Shifting Demand Out

Example: Increase in the price of a substitute



Shifting Both Curves

Example: Demand moves out and supply moves out.



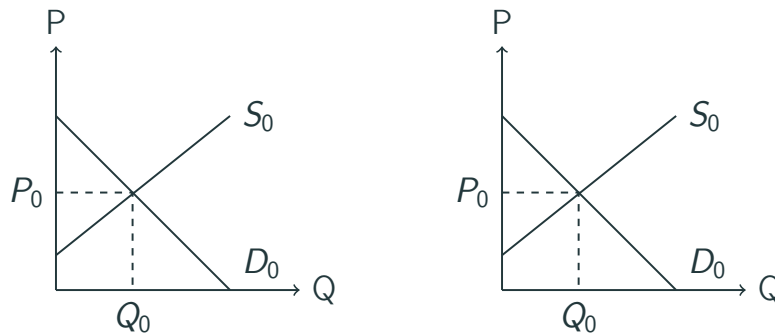
Working Through Changes

Steps

- Does it affect demand? How?
- Does it affect supply? How?
- How does the combination affect equilibrium price and quantity?

Working Backwards

Market price and quantity increase. What could have happened?



Summary

- We add up quantities supplied (demanded) at a given price to get market supply (demand).
- Market price is stable when quantity supplied is equal to quantity demanded.
- Market price is a signal of value and scarcity of a good.